

Spring 1991

Problem 1 (Sp91). List, to within isomorphism, all the finite groups whose orders do not exceed 5.

Problem 2 (Su79, Sp82, Sp91, Sp96). Let f be a continuous complex valued function on $[0, 1]$, and define the function g by

$$g(z) = \int_0^1 f(t)e^{tz} dt \quad z \in \mathbb{C}$$

Prove that g is analytic in the entire complex plane.

Problem 3 (Sp91). For n a positive integer, let $d(n)$ denote the number of positive integers that divide n . Prove that $d(n)$ is odd if and only if n is a perfect square.

Problem 4 (Sp91). Let p be a prime number and R a ring with identity containing p^2 elements. Prove that R is commutative.

Problem 5 (Sp91). Let $A = (a_{ij})_{i,j=1}^r$ be a square matrix with integer entries.

1. Prove that if an integer n is an eigenvalue of A , then n is a divisor of $\det A$, the determinant of A .
2. Suppose that n is an integer and that each row of A has sum n :

$$\sum_{j=1}^r a_{ij} = n \quad 1 \leq i \leq r$$

Prove that n is a divisor of $\det A$.

Problem 6 (Sp91). Let the vector field F in $\mathbb{R}^3$ have the form

$$F(r) = g(\|r\|)r \quad r \neq (0, 0, 0)$$

where g is a real valued smooth function on $(0, \infty)$. (F is undefined at $(0, 0, 0)$.) Prove that

$$\int_C F \cdot ds = 0$$

for any smooth closed path C in $\mathbb{R}^3$ that does not pass through the origin.

Problem 7 (Sp91). Let the function f be analytic in the unit disc, with $|f(z)| \leq 1$ and $f(0) = 0$. Assume that there is a number r in $(0, 1)$ such that $f(r) = f(-r) = 0$. Prove that

$$|f(z)| \leq |z| \left| \frac{z^2 - r^2}{1 - r^2 z^2} \right|$$

Problem 8 (Sp91). Let T be a real, symmetric, $n \times n$, *tridiagonal* matrix:

$$T = \begin{pmatrix} a_1 & b_1 & 0 & & 0 \\ b_1 & a_2 & b_2 & & \\ 0 & b_2 & a_3 & & \\ & & & 0 & \\ & & & b_{n-1} & \\ 0 & & 0 & b_{n-1} & a_n \end{pmatrix}$$

All entries not on the main diagonal or the diagonals just above and below the main one are zero. Assume $b_j \neq 0$ for all j . Prove:

1. $\text{rank } T \geq n - 1$.
2. T has n distinct eigenvalues.

Problem 9 (Sp91). Let f be a continuous function from the ball $B_n = \{x \in \mathbb{R}^n \mid \|x\| < 1\}$ into itself. Assume $\|f(x)\| < \|x\|$ for all nonzero $x \in B_n$. Let x_0 be a nonzero point of B_n , and define the sequence (x_k) by setting $x_k = f(x_{k-1})$. Prove that $\lim x_k = 0$.

Problem 10 (Sp91). Prove that $\mathbb{Q}$, the additive group of rational numbers, cannot be written as the direct sum of two nontrivial subgroups.
Note: See also ??????.

Problem 11 (Sp91). For which real numbers x does the infinite series converge?

$$\sum_{n=1}^{\infty} \frac{\sqrt{n+1} - \sqrt{n}}{n^x}$$

Problem 12 (Sp91). Let A be the set of positive integers that do not contain the digit 9 in their decimal expansions. Prove that

$$\sum_{a \in A} \frac{1}{a} < \infty$$

that is, A defines a convergent subseries of the harmonic series.

Problem 13 (Sp91). Prove that $\lim_{R \rightarrow \infty} \int_{-R}^R \frac{\sin x}{x - 3i} dx$ exists and find its value.

Problem 14 (Sp91). Let $x(t)$ be a nontrivial solution to the system

$$\frac{dx}{dt} = Ax \quad \text{where} \quad A = \begin{pmatrix} 1 & 6 & 1 \\ -4 & 4 & 11 \\ -3 & -9 & 8 \end{pmatrix}$$

Prove that $\|x(t)\|$ is an increasing function of t .

Problem 15 (Sp91). Let G be a finite nontrivial group with the property that for any two elements a and b in G different from the identity, there is an element c in G such that $b = c^{-1}ac$. Prove that G has order 2.

Problem 16 (Sp91). Let A be a linear transformation on an n -dimensional vector space over $\mathbb{C}$ with characteristic polynomial $(x - 1)^n$. Prove that A is similar to A^{-1} .

Problem 17 (Sp91). Let the function f be analytic in the punctured disc $0 < |z| < r_0$, with Laurent series

$$f(z) = \sum_{-\infty}^{\infty} c_n z^n$$

Assume there is a positive number M such that

$$r^4 \int_0^{2\pi} |f(re^{i\theta})|^2 d\theta < M \quad 0 < r < r_0$$

Prove that $c_n = 0$ for $n < -2$.

Problem 18 (Sp91). Let the real valued function f be defined in an open interval about the point a on the real line and be differentiable at a . Prove that if (x_n) is an increasing sequence and (y_n) is a decreasing sequence in the domain of f , and both sequences converge to a , then

$$\lim_{n \rightarrow \infty} \frac{f(y_n) - f(x_n)}{y_n - x_n} = f'(a)$$